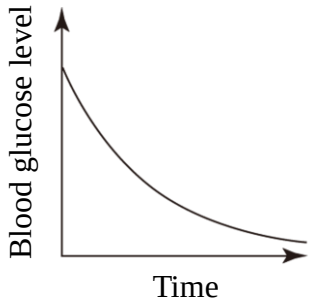
Multiple-choice questions

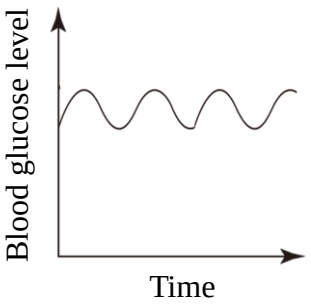
|!|BCEMB0190000001|!| ★

Which of the following graphs correctly shows the regulation of the blood glucose level through a negative feedback mechanism?

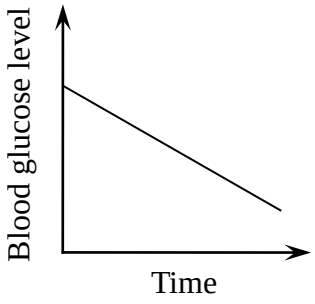
A.



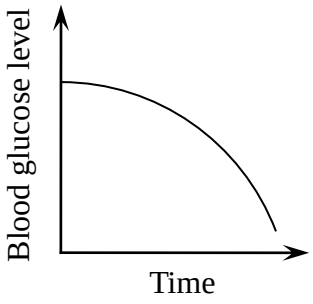
B.



C.



D.



□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0190000002|!|

Which of the following statements about the pancreas is/are ***incorrect***?

(1) It is an endocrine gland which secretes the hormones insulin and glucagon into the blood.
(2) It is an exocrine gland which secretes pancreatic juice to the pancreatic duct.
(3) It converts excess glucose to glycogen.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

□

##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000003|!|

Which of the following correctly describes the action of insulin?

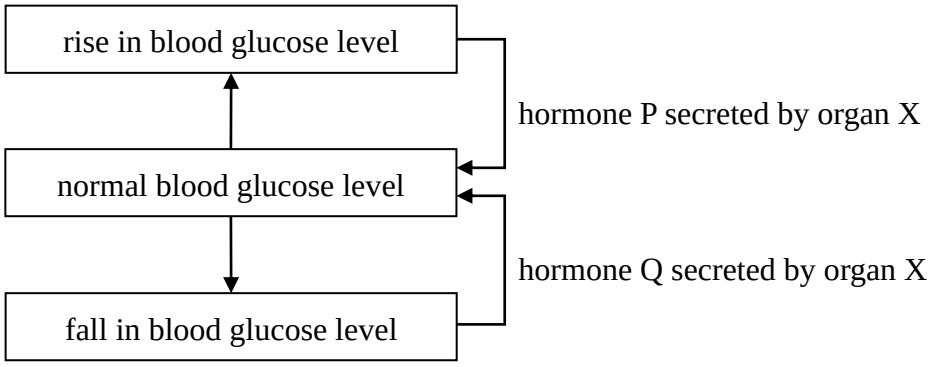
A. It catalyses the conversion of glucose to glycogen.
B. It catalyses the conversion of glycogen to glucose.
C. It stimulates liver cells to convert glucose to glycogen.
D. It stimulates liver cells to convert glycogen to glucose.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0190000004|!| ★

The diagram below shows that the blood glucose level is regulated through the actions of two hormones.



Which of the following combinations correctly identifies hormone P, hormone Q and organ X?

	Hormone P	***Hormone Q***	***Organ X***
A.	insulin	glycagon	liver
B.	glycagon	insulin	pancreas
C.	glycagon	insulin	liver
D.	insulin	glycagon	pancreas

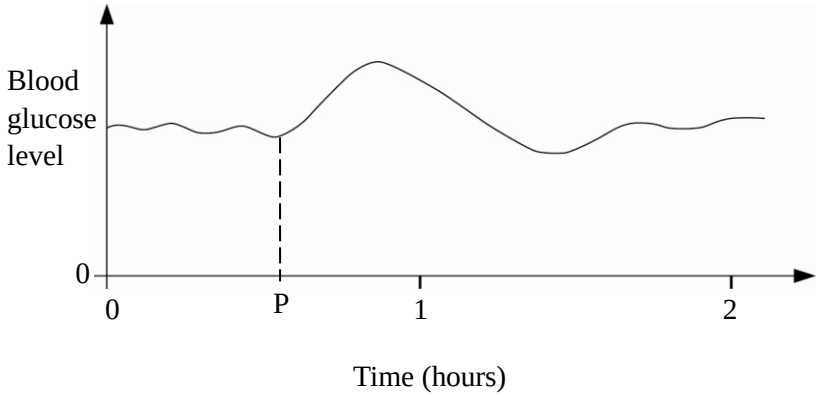
##D##

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Skill assessed	Mark
Applying concepts	1

|!|BCEMB0190000005|!| ★

The following graph shows the changes in the blood glucose level of Mr. Wong during two hours.



Which of the following explains the shape of the curve after time P?

A. Mr. Wong had an insulin injection at P.
B. Mr. Wong had been fasting since P.
C. Mr. Wong started to do physical exercise at P.
D. Mr. Wong ate a meal rich in carbohydrates at P.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0190000006|!| ★

Which of the following statements about insulin and glucagon is correct?

A. Insulin is produced by the liver.
B. Glucagon is an enzyme which catalyses the conversion of glycogen to glucose.
C. The level of insulin in blood is higher before a meal.
D. During fasting, the level of glucagon in blood increases.

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0190000007|!|

Which of the following statements about glucagon is correct?

A. Glucagon is secreted by the liver.
B. Glucagon is an enzyme which catalyses the conversion of glycogen into glucose.
C. The level of glucagon in blood is higher after a meal rich in carbohydrates.
D. During fasting, the level of glucagon in blood increases.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000008|!|

Some diabetic patients receive insulin by injection rather than orally because

A. insulin would be digested if it were taken orally.
B. insulin cannot be absorbed through the alimentary canal.
C. the insulin injected can exert effect more stably.
D. insulin injection is cheaper than oral insulin pills.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000009|!|

Insulin is secreted from the

A. gall bladder.
B. skeletal muscles.
C. liver.
D. pancreas.

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0190000010|!|

Glucagon is secreted from the

A. pancreas.
B. pancreatic duct.
C. liver.
D. pituitary gland.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000011|!|

Which of the following is/are kept constant in the human body?

(1) body temperature
(2) insulin concentration in blood
(3) carbon dioxide content in blood

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000012|!|

Which of the following will occur if insulin is injected into a person?

A. Body cells take up more glucose from the blood.
B. More glycogen is converted to glucose.
C. More glucose is excreted in urine.
D. The pancreas produces more insulin.

##A##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0190000013|!|

Which of the following statements about insulin is ***incorrect***?

A. Insulin is protein in nature.
B. Insulin is secreted by the pancreas.
C. More insulin is secreted when there is a fall in the blood glucose level.
D. Insulin stimulates the conversion of glucose to glycogen by liver cells.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000014|!|

Which of the following action(s) is/are exerted by insulin?

(1) stimulating liver cells to convert glucose to glycogen for storage
(2) stimulating body cells to take up more glucose
(3) inhibiting the absorption of glucose in the intestines

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000015|!|

Which of the following is ***not*** the cause of low blood glucose level in diabetics?

A. injecting too much insulin
B. exercising too vigorously
C. fasting
D. eating a meal rich in starch

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0190000016|!|

How would the blood glucose level and blood insulin level of a normal person change within five minutes after the ingestion of 200 cm^3 of concentrated glucose solution?

	Blood glucose level	***Blood insulin level***
A.	increase	decrease
B.	increase	increase
C.	decrease	decrease
D.	decrease	increase

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0190000017|!|

Which of the following statements about insulin is correct?

A. Its secretion increases in response to a drop in blood glucose level.
B. It is secreted by the liver.
C. It catalyses the conversion of glucose to glycogen in the liver.
D. It is composed of amino acids.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000018|!|

Which of the following occur(s) during exercise?

(1) The secretion of glucagon decreases.
(2) The secretion of insulin increases.
(3) The rate of breathing increases.

A. (3) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0190000019|!|

Which of the following substances is/are kept at a constant level in the blood of humans?

(1) glucose
(2) glucagon
(3) salt

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000020|!|

Which of the following functions is/are performed by the human kidneys?

(1) osmoregulation
(2) thermoregulation
(3) regulation of blood glucose level

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0190000021|!|

Which of the following parameters of our internal environment is/are regulated by a negative feedback mechanism?

(1) body temperature
(2) blood glucose level
(3) water content in blood

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000022|!|

The following events are involved in the regulation of blood glucose level in the human body.

(1) The change in the blood glucose level is detected by the pancreas.
(2) The blood glucose level falls.
(3) Liver cells are stimulated to convert excess glucose to glycogen.
(4) The blood glucose level increases.
(5) More insulin and less glucagon are secreted into the blood.

The correct order should be:

A. (2) → (1) → (5) → (3) → (4)
B. (4) → (1) → (5) → (3) → (2)
C. (4) → (3) → (1) → (5) → (2)
D. (4) → (5) → (1) → (3) → (2)

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0190000023|!| ★

The presence of glucose in urine is ***most*** probably caused by

A. eating a meal which is rich in carbohydrates.
B. too much glycogen in the liver.
C. an increased absorption of glucose in the small intestine.
D. inadequate secretion of insulin by the pancreas.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0190000024|!| ★

Which of the following statements about diabetic patients is ***incorrect***?

A. Their blood glucagon level remains at a high level.
B. Glucose is excreted in their urine.
C. Their blood insulin level remains at a low level.
D. The amount of glycogen stored in the liver remains low.

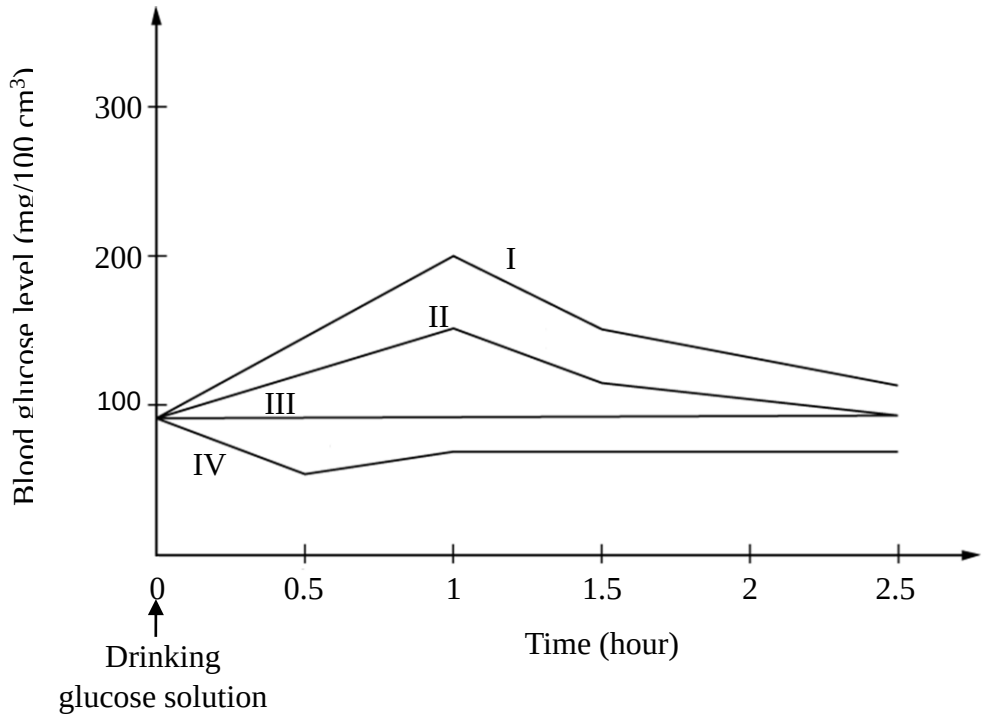
##A##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0190000025|!| ★

In an investigation, a healthy man and a diabetic patient were asked to drink a glass of glucose solution. Their blood glucose levels were then measured at 30-minute intervals for 2.5 hours. Two of the curves in the graph below show the results.



Which curves shows the changes in the blood glucose level of the healthy man and the diabetic patient after drinking the glucose solution?

	Healthy man	***Diabetic patient***
A.	I	III
B.	II	I
C.	II	IV
D.	III	II

□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0190000026|!|

Which of the following are the functions of insulin?

(1) to increase the uptake of glucose by body cells
(2) to break down the glucose in blood
(3) to enhance the conversion of glucose to glycogen in the liver

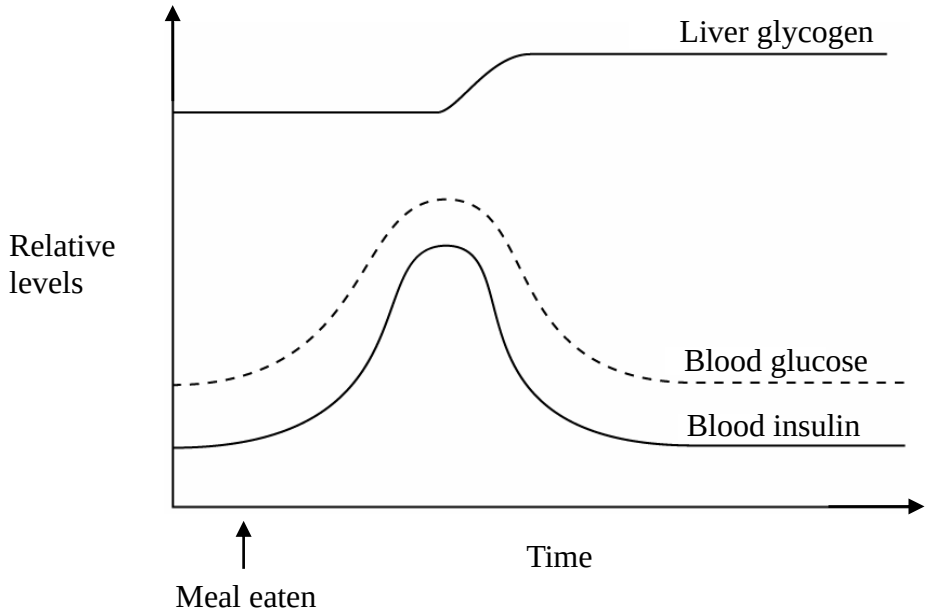
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0190000027|!| ★

The following graph shows the relative levels of liver glycogen, blood glucose and blood insulin of a man.



Which of the following statements about the graph is/are correct?

(1) The liver glycogen level increases as the blood insulin level decreases.
(2) The blood glucose level decreases because insulin stimulates the conversion of glucose to glycogen.
(3) The blood glucose level is directly proportional to the blood insulin level.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (2) and (3) only

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0190000028|!|

Insulin is carried to the target organ through

A. the pancreatic duct.
B. the lymph.
C. the plasma.
D. the blood cells.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000029|!|

Which of the following combinations shows the organs that are involved in the regulation of body temperature and water content in blood respectively?

	Regulation of body temperature	***Regulation of water content in blood***
A.	kidneys	skin
B.	skin	kidneys
C.	skin	liver
D.	liver	skin

##B##

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Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0190000030|!|

The oxygen and carbon dioxide content in blood is maintained constant by controlling

(1) the rate of breathing.
(2) the heart rate.
(3) the rate of sweating.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

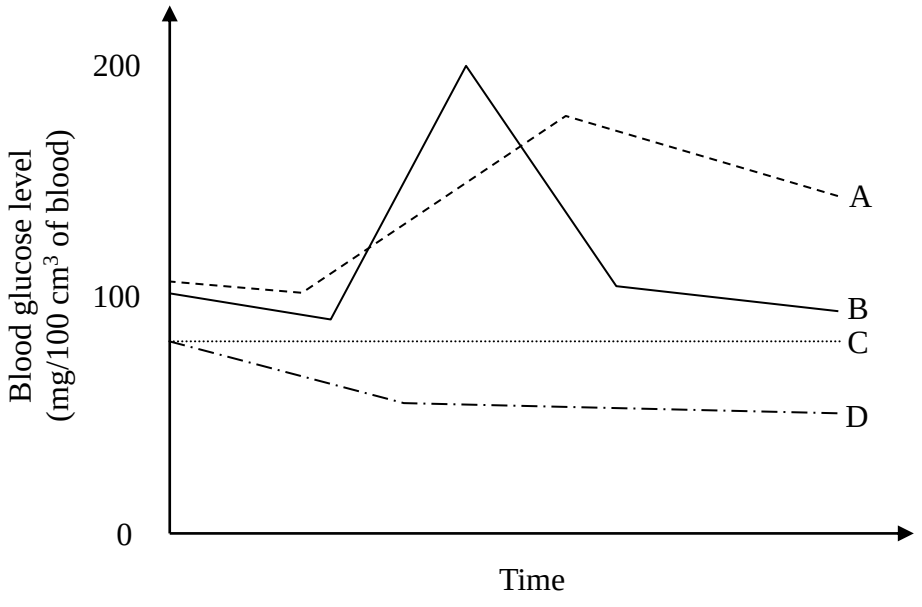
□

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0190000032|!| ★

Directions: The following three questions refer to the graph below which shows the blood glucose level of four patients from a hospital.



Which patients suffer from diabetes mellitus?

A. patient A
B. patient B
C. patient C
D. patient D

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0190000033|!| ★

Which of the following might be the probable reason(s) that lead(s) to the low blood glucose level in patient D?

(1) fasting
(2) no response of body cells to glucagon
(3) over-secretion of insulin by the pancreas

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0190000034|!| ★

What causes the drop in the blood glucose level in patient B?

A. an increase in the blood glucagon level
B. an increase in the blood insulin level
C. a decrease in the uptake of glucose by body cells
D. a decrease in the rate of respiration in liver cells

##B##

Skill assessed	Mark
Applying concepts	1

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